Out-of-Lab Report: Cavendish Torsion Balance Measurement of the Gravitational Constant

Hongyu Wang
Partner(s): Cici Zhang

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Abstract

We used a PASCO AP-8215A torsion balance to measure the gravitational constant G by converting the tiny gravitational torque between large source masses and small suspended masses into an amplified laser-spot displacement on a distant scale. Our primary result came from PASCO Method II (equilibrium centers) applied to two in-lab long runs. Averaging the two Method II determinations gives

$$G = (3.12 \pm 0.29_{\text{stat}} \pm 0.52_{\text{run}}) \times 10^{-10}\ \text{N}\,\text{m}^2\,\text{kg}^{-2},$$

which differs from the accepted value by 367%. A secondary Method III (acceleration) analysis gave $G = (3.94 \pm 0.08) \times 10^{-10}\ \text{N}\,\text{m}^2\,\text{kg}^{-2}$. Method I (final deflection) could not be executed as prescribed: even after leaving the apparatus undisturbed overnight ($\sim$1 day), the spot continued to oscillate and drift, so resting endpoints S_1, S_2 were not observable. We therefore report a *Method-I simulation* using the run with the largest excursions, defining "endpoints" by robust distribution envelopes (1%–99% quantiles), yielding $\Delta S_{\text{M1,sim}} \approx 60.7$ cm (not a valid G measurement, but a stress test). The largest uncertainties arose from alignment sensitivity, possible sphere–plate contact, source-mass placement asymmetry of about 2 mm, and reading/calibrating the laser-spot position.

1 Introduction

The goal of this experiment was to measure Newton's gravitational constant G by observing the torsional response of a suspended two-mass system. When the large source masses are brought near the small masses on the torsion balance, the mutual gravitational attraction exerts a torque on the horizontal rod. The suspension fiber provides a restoring torque, so the apparatus behaves as both a static torque balance and a torsional oscillator.

This experiment is powerful because the gravitational force between laboratory masses is extremely small. Using the accepted value of G and our apparatus geometry (Table 3), the expected near-force scale is

$$F_{\text{sig}} = G\frac{m_1 m_2}{b^2} \approx 2.15e-09\ \text{N}. \tag{1}$$

A torsion balance converts this tiny force into an angular deflection, and the mirror-plus-laser optical lever converts that small angle into a centimeter-scale spot shift that can be measured on a distant ruler. Thus the apparatus makes it possible to infer an extremely weak interaction from a much larger optical displacement.

In practice, the experiment does not "measure G directly". Instead it measures three things: the torsional period T, the equilibrium shift or early-time acceleration of the laser spot when the source masses are moved, and the geometry that links optical displacement to torque. The purpose of this report is therefore twofold: to calculate G, and to explain which experimental realities most strongly controlled the quality of that calculation.

2 Theory

Optical lever

When the source masses are switched, the equilibrium twist changes from $-\theta$ to $+\theta$. Because the angle doubles upon reflection, the laser beam direction changes by 4θ and the spot shift on a scale a distance L away is

$$\theta = \frac{\Delta S}{4L}, \qquad \Delta S \equiv S_2 - S_1. \tag{2}$$

Method I and Method II

For the deflection-based methods, PASCO gives

$$G = \frac{\pi^2 \Delta S\, b^2}{m_1 L d} \frac{\left(d^2 + \frac{2}{5} r^2\right)}{T^2}, \tag{3}$$

where b is the source-to-test center separation, d is the lever arm to each small mass, r is the small-mass radius, and T is the torsional period. The manual also notes that the far source mass weakly modifies the net torque, so the corrected value is

$$\beta = \frac{b^3}{(b^2 + 4d^2)^{3/2}}, \qquad G_0 = \frac{G}{1-\beta}. \tag{4}$$

Method III

Immediately after switching the source masses, the first part of the motion may be treated approximately as constant acceleration. Then

$$G = \frac{b^2 a_0}{2m_1}, \qquad a_0 = \left(\frac{\Delta S}{t^2}\right) \frac{d}{L}, \tag{5}$$

where the slope of ΔS versus t^2 gives the initial acceleration.

Expected signal scale

Inserting the accepted value $G_{\mathrm{acc}} = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ into Eq. (3) gives an expected optical shift

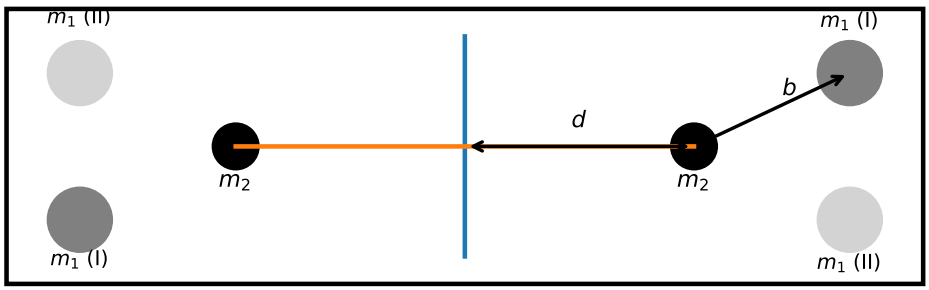
$$\Delta S_{\mathrm{exp}} = \frac{G_{\mathrm{acc}}(1-\beta) m_1 L d}{\pi^2 b^2} \frac{T^2}{\left(d^2 + \frac{2}{5} r^2\right)} \approx 2.57\,\mathrm{cm}. \tag{6}$$

This number is useful as a reality check: if the observed ΔS is much larger than ΔS_{exp}, the excess must be coming from non-ideal mechanics, geometry, or readout rather than from gravity alone.

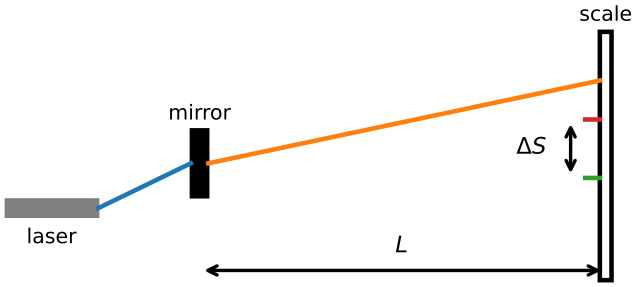
3 Methods

3.1 Apparatus and operating principle

The apparatus was a PASCO AP-8215A gravitational torsion balance containing two small test masses suspended from a thin torsion ribbon and two large source masses that can be moved between two repeatable positions. A mirror mounted on the pendulum reflected a laser beam onto a distant scale. The top-view and side-view geometry used in the analysis are shown in Fig. 1. The diagram indicates which components actually enter the calculation: the lever arm d, the source-mass separation b, the optical-lever length L, and the spot shift ΔS.



(a) Top view of the torsion-balance geometry. The large source masses exert gravitational forces on the small suspended masses and produce a torque about the fiber.



(b) Side view of the optical lever. Small mirror rotations are amplified into a measurable spot displacement ΔS on a scale at distance L.

Figure 1: Schematic of the apparatus and the geometric quantities used to calculate G.

3.2 Detailed alignment procedure

The setup process was the most difficult part of the experiment. Following the manual, the base was leveled with the built-in leveling feet while viewing through the leveling sight. The working criterion was that the reflection of the pendulum bob in the mirror appeared centered. In practice this was extremely sensitive: a tiny change in one leveling foot changed the apparent up/down symmetry and shifted the laser spot noticeably.

The pendulum height was then adjusted so that the assembly could move freely without touching the case. This was essential because any contact between a small sphere (or the pendulum arm) and a chamber plate would add non-gravitational restoring forces. Our lab notes show that we

spent roughly ten hours across several sessions trying to find a stable balance point. To isolate the mechanical zero without gravitational bias from the source masses, we temporarily removed the large masses and adjusted the apparatus until the red spot oscillated with approximately equal left and right amplitudes about a common midpoint. Only after this criterion was met did we reinstall the source masses and begin the formal runs.

3.3 Source-mass placement problem

A second setup difficulty was source-mass reproducibility. The ideal model assumes both large masses sit in a symmetric geometry when moved to Position I or II. In practice we found that when one large mass appeared fully seated, the opposite mass could still retain a visible gap. Using a paper-stack gauge calibrated with a caliper, we estimated this residual asymmetry to be about 2 mm. Because the gravitational force depends on $1/b^2$, this geometric imperfection was carried forward as an explicit systematic uncertainty rather than ignored as a vague setup issue.

3.4 Measurement procedures

For Method I, the apparatus was left undisturbed overnight in an attempt to read static endpoints S_1 and S_2. This did not succeed: the spot still oscillated and drifted the next day, so Method I could not be used as intended.

For Method II, long videos were recorded before and after switching the source masses. The resulting trajectories were fit to damped oscillations, and the equilibrium centerlines were taken as S_1 and S_2. This method was chosen as the primary analysis because it does not require the system to come completely to rest.

For Method III, the first part of the post-switch motion was analyzed as approximate constant-acceleration motion. This gives an independent result, but it is much more sensitive to the exact start time and to any transient mechanical disturbance.

3.5 Video tracking and calibration

The laser spot was recorded on video and tracked with an OpenCV centroid detector. Two tape markers separated by 100 cm on the ruler were used for pixel-to-centimeter calibration. A reading uncertainty of about 0.10 cm was assigned to the spot position to account for centroid noise, ruler reading, and calibration uncertainty.

4 Raw Data and Analysis

4.1 Period measurements over multiple cycles

Because the in-lab traces showed abrupt high-frequency bursts that were likely caused by intermittent contact, the cleanest torsional period estimate came from a cleaner reference trace processed with the same tracking code. Table 1 lists peak-to-peak measurements from that trace. Two disturbed cycles were excluded before averaging.

Table 1: Peak-to-peak period measurements from a clean reference trace.

Peak 1 (s)	Peak 2 (s)	Single period (s)	Status
3460.9	3751.9	291.0	used
3751.9	4079.5	327.6	excluded
4079.5	4372.6	293.1	used
4372.6	4676.7	304.1	used
4676.7	4980.4	303.7	used
4980.4	5304.4	324.0	excluded
5304.4	5595.1	290.7	used
5595.1	5886.8	291.7	used

The peak-picking average was

$$T_{\text{raw}} = (295.7 \pm 6.4)\,\text{s},$$

consistent with the fit-derived value

$$T = (292.3 \pm 12.8)\,\text{s}$$

used in the final calculations. Measuring multiple cycles and averaging is preferable to timing only one oscillation because it reduces the relative timing uncertainty and exposes obviously disturbed intervals.

4.2 Laser spot positions and equilibrium shifts

Table 2 lists the equilibrium positions obtained from the damped-oscillation fits. The two in-lab runs are the basis of the primary reported value, while the reference run is included mainly as a pipeline sanity check and to support the period estimate.

Table 2: Equilibrium positions extracted from fitted centerlines.

Run	S_1 (cm)	S_2 (cm)	ΔS (cm)
Reference trace	51.69	57.60	5.908
In-lab alt/full	93.11	103.73	10.618
In-lab main/100min	82.61	96.03	13.427

4.3 Geometric parameters

The parameters entering the calculation are collected in Table 3. The last row records the observed source-mass asymmetry, since that offset is more informative than quoting only an ideal nominal separation.

Table 3: Measured and nominal parameters entering the calculation of G.

Quantity	Value	Comment
Source mass m_1	(1.500 ± 0.010) kg	manual nominal
Test mass m_2	(38.3 ± 0.2) g	manual nominal
Lever arm d	0.0500 m	manual nominal
Small-mass radius r	0.008 19 m	manual nominal
Mirror-to-scale distance L	(2.838 ± 0.010) m	measured
Nominal center separation b	(0.0422 ± 0.0006) m	used in calculation
Placement asymmetry δb	(2.0 ± 0.5) mm	one mass seated, opposite gap visible

4.4 Sample calculations

Method I (simulated envelope, main run). To illustrate why Method I fails without true settling, we use one simulated endpoint pair from the main run envelope:

$$S_{1,\mathrm{sim}} = 60.97\,\mathrm{cm}, \qquad S_{2,\mathrm{sim}} = 121.70\,\mathrm{cm},$$

so that

$$\Delta S_{\mathrm{sim}} = S_{2,\mathrm{sim}} - S_{1,\mathrm{sim}} = 60.73\,\mathrm{cm} = 0.6073\,\mathrm{m}.$$

Substituting this into Eq. (3),

$$G = \frac{\pi^2(0.6073)(0.0422)^2}{(1.500)(2.838)(0.0500)} \frac{0.002527}{(292.3)^2} = 1.48 \times 10^{-9}\ \mathrm{N\,m^2\,kg^{-2}},$$

and after the far-mass correction,

$$G_0 = \frac{G}{1-\beta} = 1.58 \times 10^{-9}\ \mathrm{N\,m^2\,kg^{-2}}.$$

This value is not reported as a physical measurement; it quantifies how badly Method I can overestimate G if one reads ongoing oscillation extremes as though they were resting endpoints.

Method II (main in-lab run). For the main run, Eq. (3) gives

$$\Delta S = 13.427\,\mathrm{cm} = 0.134\,27\,\mathrm{m}, \qquad d^2 + \frac{2}{5}r^2 = 0.002\,527\,\mathrm{m}^2.$$

Therefore

$$G = \frac{\pi^2(0.13427)(0.0422)^2}{(1.500)(2.838)(0.0500)} \frac{0.002527}{(292.3)^2} = 3.28 \times 10^{-10}\ \mathrm{N\,m^2\,kg^{-2}},$$

and after the far-mass correction,

$$G_0 = \frac{G}{1-\beta} = 3.48 \times 10^{-10}\ \mathrm{N\,m^2\,kg^{-2}}.$$

Method III. For the worksheet dataset, the fitted early-time slope was

$$k = \frac{\Delta S}{t^2} = 0.0035\,\mathrm{cm/s^2}.$$

Using Eq. (5),

$$a_0 = k(0.01)\frac{d}{L} = 6.25 \times 10^{-7}\ \mathrm{m/s^2},$$

so

$$G_0 = (3.94 \pm 0.08) \times 10^{-10}\ \mathrm{N\,m^2\,kg^{-2}}.$$

Primary reported value from the two in-lab Method II runs. Our primary value is the average of the two in-lab Method II determinations:

$$\bar{G}_{\mathrm{II}} = \frac{G_{\mathrm{alt}} + G_{\mathrm{main}}}{2} = 3.12 \times 10^{-10}\ \mathrm{N\,m^2\,kg^{-2}}.$$

The run-to-run reproducibility uncertainty is estimated by the sample standard deviation,

$$u_{\mathrm{run}} = \sqrt{\frac{\sum_i (G_i - \bar{G})^2}{n-1}} = 5.15 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}},$$

while the propagated statistical component is

$$u_{\mathrm{stat}} = 2.94 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}.$$

Combining the two in quadrature gives

$$u_{\mathrm{tot}} = 5.93 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}.$$

5 Results

Our primary result is therefore

$$G = (3.12 \pm 0.29_{\mathrm{stat}} \pm 0.52_{\mathrm{run}}) \times 10^{-10}\ \mathrm{N\,m^2\,kg^{-2}},$$

with percent error

$$\%\,\mathrm{error} = \left| \frac{G_{\mathrm{meas}} - G_{\mathrm{acc}}}{G_{\mathrm{acc}}} \right| \times 100\% \approx 367\%.$$

Table 4: Comparison of the three PASCO methods.

Method	**Data used**	G_0 $(\mathrm{N\,m^2\,kg^{-2}})$	**Percent error**	**Note**
I (sim.)	main-run envelope	1.58×10^{-9}	—	Simulated; no true settling.
II (alt/full)	in-lab long run	2.76×10^{-10}	313%	Long-run fit.
II (main/100min)	in-lab long run	3.48×10^{-10}	422%	Long-run fit.
II (primary)	**mean of two in-lab runs**	3.12×10^{-10}	**367%**	**Reported value.**
III	worksheet short-time data	3.94×10^{-10}	490%	Timing-sensitive.

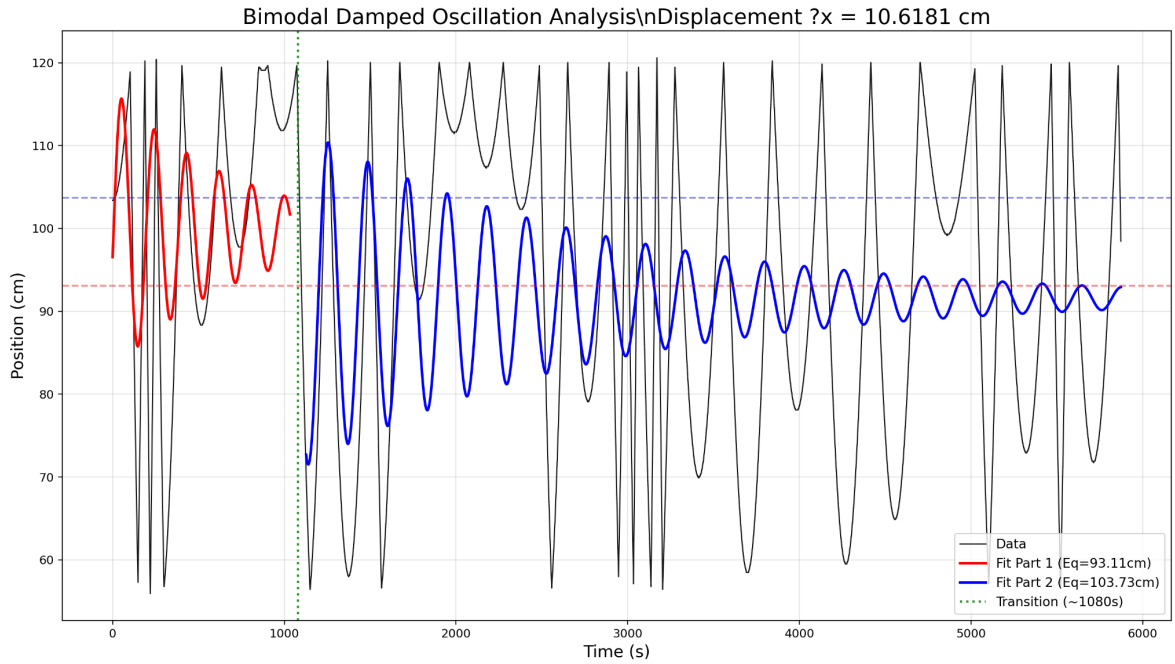


Figure 2: Method II fit for the in-lab alt/full run. The fitted centerlines yield $\Delta S = 10.618\,\mathrm{cm}$.

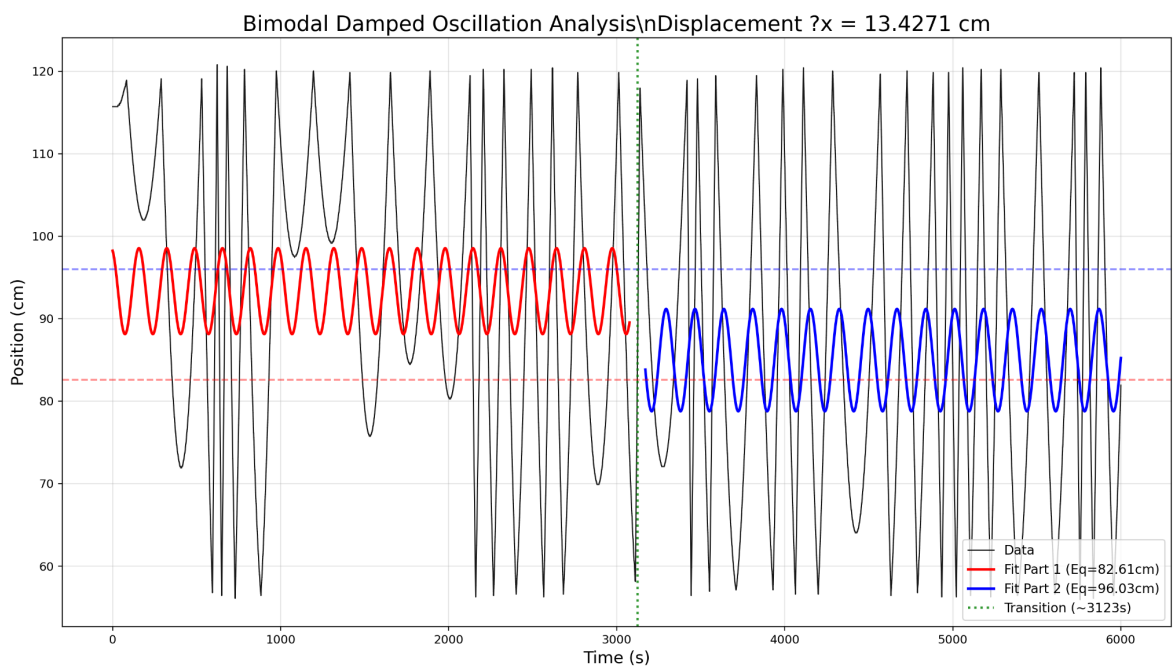


Figure 3: Method II fit for the in-lab main/100min run. The fitted centerlines yield $\Delta S = 13.427\,\mathrm{cm}$.

6 Discussion

Why the result is of the right order, but not accurate

The measured values are all of the correct order of magnitude for a Cavendish experiment, which means the apparatus did respond to the gravitational reconfiguration of the source masses. However, the primary in-lab value is still 367% above the accepted value. That discrepancy is much larger than the formal statistical uncertainty, so the limiting issue in this experiment was not random timing noise or one more significant figure in the algebra. The dominant issue was whether the apparatus actually behaved like the ideal torsion balance assumed in the derivation.

Statistical uncertainty versus real experimental uncertainty

For Methods I and II, the propagated statistical uncertainty is

$$\left(\frac{u_G}{G}\right)^2 \approx \left(\frac{u_{\Delta S}}{\Delta S}\right)^2 + \left(2\frac{u_b}{b}\right)^2 + \left(\frac{u_L}{L}\right)^2 + \left(2\frac{u_T}{T}\right)^2 + \left(\frac{u_{m_1}}{m_1}\right)^2. \tag{7}$$

For the main run, the individual contributions are

$$\frac{u_{\Delta S}}{\Delta S} = 1.67\%, \quad 2\frac{u_b}{b} = 2.84\%, \quad \frac{u_L}{L} = 0.35\%, \quad 2\frac{u_T}{T} = 8.76\%, \quad \frac{u_{m_1}}{m_1} = 0.67\%.$$

These combine to a statistical fractional uncertainty of only about 9.4%. But the two in-lab Method II runs differ by much more than that, which is why the run-to-run scatter had to be reported separately. This is exactly what one expects when systematic effects dominate the experiment.

Specific error sources and how they influence G

Alignment difficulty. The fact that the system was so sensitive to minute leveling changes already indicated that the apparatus was close to a mechanically delicate condition. A balance point that is hard to reproduce is also hard to trust. This difficulty is why the start criterion had to be based on equal left/right amplitudes rather than on a truly motionless spot.

Possible sphere–plate contact. This was the most serious issue. Abrupt high-frequency bursts in the time series and a shortened apparent period strongly suggest that the pendulum occasionally touched a plate. Because Methods I and II use $G \propto 1/T^2$, a shorter apparent period pushes the inferred G upward. If the true free period were about 600 s but the measured period were 292.3 s, then

$$\frac{G_{\text{meas}}}{G_{\text{true}}} = \left(\frac{T_{\text{free}}}{T_{\text{meas}}}\right)^2 \approx 4.21,$$

which is a 321% high bias. This is large enough to explain why our measured values are several times the accepted constant. Thus contact is not a minor source of noise; it is a plausible dominant distortion of the apparatus dynamics.

Source-mass asymmetry. If one source mass is fully seated while the other is farther away by $\delta b \approx 2\,\text{mm}$, the torque coefficient is not exactly $2/b^2$. Instead,

$$\frac{1}{b_1^2} + \frac{1}{b_2^2} = \frac{2}{b^2} f, \qquad f = \frac{1}{2}\left[1 + \left(\frac{b}{b+\delta b}\right)^2\right].$$

For $b = 42.2\,\mathrm{mm}$ and $\delta b = 2\,\mathrm{mm}$, $f \approx 0.956$. Ignoring the asymmetry therefore biases the inferred G low by about 4.4%. This is a real geometric error, but its sign is opposite to the contact-induced period shortening, so different systematics can partially mask one another.

Laser-spot reading and calibration. Because $G \propto \Delta S$, any fractional error in the spot shift becomes the same fractional error in G. A centroid uncertainty of about one pixel corresponds to about 0.06 cm for the 100 cm / 1625 px calibration used here, already a few percent of the expected signal. This makes optical readout quality much more important than in a less sensitive apparatus.

Quantified systematic effects

Table 5 converts several realistic disturbances into an equivalent spurious $\delta(\Delta S)$. Since $G \propto \Delta S$ in Methods I and II, the same fraction approximately gives $\delta G/G$.

Table 5: Estimated systematic effects expressed as equivalent spurious $\delta(\Delta S)$.

Effect	Key relation	$\delta(\Delta S)$ (cm)	$\delta G/G$	Direction
70 kg person at 1 m	$\delta(\Delta S) \approx (\tau_{\mathrm{person}}/\tau_{\mathrm{sig}})\Delta S_{\mathrm{exp}}$	0.11	4.2%	$\pm$
Person moves 1 m to 2 m	same, using $\Delta F = F(1) - F(2)$	0.08	3.1%	$\pm$
Base tilt 0.1 mrad	$\delta S \approx 4L\delta\theta$	0.11	4.4%	$\pm$
Ribbon drift (example)	direct shift over hours	0.50	19.4%	$\pm$
Source-mass asymmetry	$\delta G/G \approx -(1-f)$	0.11	4.4%	low
Finite-twist geometry	$\delta F/F \approx 2(d\theta)/b$	0.01	0.5%	high
Scale calibration example	$\delta(\Delta S)/\Delta S \sim \delta s/s$	0.01	0.4%	$\pm$
Sphere-plate contact	$\delta G/G \approx (T_{\mathrm{free}}/T)^2 - 1$	8.27	321%	high

Excluding the deliberately extreme contact stress case, a representative systematic scale is

$$u_{\mathrm{sys}}(\Delta S) \approx \sqrt{\sum_i \delta(\Delta S)_i^2} \approx 0.53\,\mathrm{cm}, \qquad \frac{u_{\mathrm{sys}}(G)}{G} \approx \frac{u_{\mathrm{sys}}(\Delta S)}{\Delta S_{\mathrm{exp}}} \approx 20.7\%.$$

This is consistent with the observation that reproducibility, not statistical precision, is the limiting factor.

What cancels, and what does not

A truly constant background torque can cancel to first order in $\Delta S = S_2 - S_1$, because it shifts both equilibria in the same direction. Some common-mode laser drift can cancel for the same reason. However, moving masses in the room, asymmetric base tilt during mass switching, and intermittent contact do not cancel automatically. The far-mass correction increases G by about 6.2%, while the observed source-mass asymmetry would bias G low by about 4.4%. The danger is therefore not just one bad correction; it is that several uncontrolled effects can partly hide one another.

How the experiment could be improved

A stronger version of this experiment would focus first on mechanical integrity. The pendulum must be guaranteed never to touch the chamber plates, ideally by direct clearance checks or modified stops. The source masses should be positioned with much better repeatability, and the actual

center-to-center distance should be measured after every move rather than inferred from an ideal nominal value.

The environment must also be controlled. Because gravitational perturbations cannot be shielded, nearby people, carts, and door motion must simply be kept away. A rigid common base for the laser, scale, and torsion balance would reduce relative tilt. Symmetric sequences such as I→II→I should be used so drift can be estimated and subtracted. Finally, the analysis should include an explicit drift term,

$$S(t) = S_0 + vt + Ae^{-t/\tau} \sin(\omega t + \phi),$$

so slow baseline motion is separated from the oscillation itself rather than absorbed into ΔS.

7 Conclusion

This experiment used a torsion balance to infer the gravitational constant from a combination of oscillation period, laser-spot displacement, and apparatus geometry. Our primary in-lab result, obtained from the average of two Method II determinations, was

$$G = (3.12 \pm 0.29_{\text{stat}} \pm 0.52_{\text{run}}) \times 10^{-10}\ \text{N}\,\text{m}^2\,\text{kg}^{-2},$$

which is 367% above the accepted value. Method III gave a similarly high value, while Method I could not be completed because the system never reached static equilibrium even after overnight settling.

The main lesson of this experiment is that the challenge is not the algebraic derivation of G but the physical validity of the torsion-balance motion. Our alignment history, the use of equal left/right amplitudes as a practical start criterion, the later appearance of abrupt bursts, and the observed source-mass gap all show that the apparatus did not remain in the ideal free-oscillation geometry assumed by the theory. Even so, the experiment successfully demonstrated the central idea of the Cavendish balance: an extremely weak gravitational interaction can be inferred from a carefully amplified angular motion. The numerical inaccuracy of our final result is therefore still scientifically useful, because it identifies exactly where the apparatus is most vulnerable - alignment, contact, geometry, and environmental control.

A Selected raw data excerpt

Table 6 lists a short excerpt of the calibrated position data from the main in-lab run at 15 s intervals. Complete frame-level tracking CSV files are included in the accompanying project archive.

Table 6: Selected raw position data excerpt from the main in-lab run.

Time (s)	Position (cm)	Assigned reading uncertainty (cm)
0.0	64.33	0.10
15.0	64.33	0.10
30.0	64.33	0.10
45.0	63.99	0.10
60.0	63.15	0.10
75.0	62.14	0.10
90.0	63.32	0.10
105.0	67.01	0.10
120.0	70.36	0.10

References

PASCO Scientific, *Gravitational Torsion Balance AP-8215A: Instruction Manual with Experiment Guide and Teachers' Notes* (012-11032B).